

# 4339 Asset Pricing and Investments - Assignment 2

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## 1 The CAPM

Consider an economy with a portfolio of risky assets and a risk-free asset in which the CAPM holds. The following table gives the expected return  $\mathbb{E}[r]$  and the standard deviation  $\sigma$  of these two assets.

| Portfolio           | Expected Return | Standard Deviation |
|---------------------|-----------------|--------------------|
| Risk free ( $r_f$ ) | 2%              | 0%                 |
| Market ( $r_M$ )    | 8%              | 20%                |

### 1.1 Computation of Beta $\beta$ for an asset that correlates with the market.

Assume asset  $A$  has standard deviation  $\sigma_A = 10\%$  and correlation  $\rho_{A,M} = 0.5$  with the market. Beta measures the sensitivity of asset  $A$ 's returns relative to the market  $M$ . It is defined as

$$\beta_A = \frac{\sigma_{A,M}}{\sigma_M^2} \quad \text{where} \quad \sigma_{A,M} = \rho_{A,M} \sigma_A \sigma_M$$

Substituting the values yields

$$\beta_A = \frac{0.5 \cdot 0.1 \cdot 0.2}{0.2^2} = 0.25.$$

Asset  $A$  carries one quarter of the market's systematic risk.

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### 1.2 Deriving the risk-aversion coefficient under the CAPM

Under the CAPM all investors hold a combination of the risk-free asset and the tangency portfolio  $T$ , which is equal to the market portfolio. Therefore,  $\mathbb{E}[r_T] = \mathbb{E}[r_M] = 8\%$  and  $\sigma_T = \sigma_M = 20\%$ .

The investor puts weight  $w_M$  into the market portfolio and weight  $1 - w_M$  into the risk-free asset, with the goal to maximize his utility function

$$U = r_f + w_M(\mathbb{E}[r_M] - r_f) - \frac{a}{2}(w_M)^2 \sigma_M^2. \quad (1)$$

Since the risk-free asset has zero variance, the standard deviation of the investor's portfolio is

$$\begin{aligned}\sigma_P &= w_M \sigma_M \\ w_M &= \frac{\sigma_P}{\sigma_M}.\end{aligned}$$

Given that the portfolio standard deviation is  $\sigma_P = 0.1$ , this results in

$$w_M = \frac{0.10}{0.20} = 0.5 \quad \text{and} \quad w_f = 1 - w_M = 0.5.$$

To find the maximum of the utility formula we differentiate (1) with respect to  $w_M$  and set it equal to zero:

$$\begin{aligned}\frac{dU}{dw_M} &= (\mathbb{E}[r_M] - r_f) - a w_M \sigma_M^2 = 0 \\ a &= \frac{(\mathbb{E}[r_M] - r_f)}{w_M \sigma_M^2}\end{aligned}$$

Substituting the values yields

$$a = \frac{0.08 - 0.02}{0.5 \cdot 0.2^2} = \frac{0.06}{0.02} = 3.$$

The investor's risk-aversion coefficient is 3.

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